connected in series with Anode (A). The working of SCR can be studied under the following two conditions.

1. ***When Gate is Open***

   Figure 6.16 shows, the forward biasing of SCR (Anode kept positive with respect to cathode). Here no gate voltage ($V_G = 0$) is applied. The supply voltage V forward biases the junction J_1 and J_3 and reverse biases the junction J_2.So no current flows through SCR. Therefore the SCR is in 'OFF' state. It offers very high resistance.

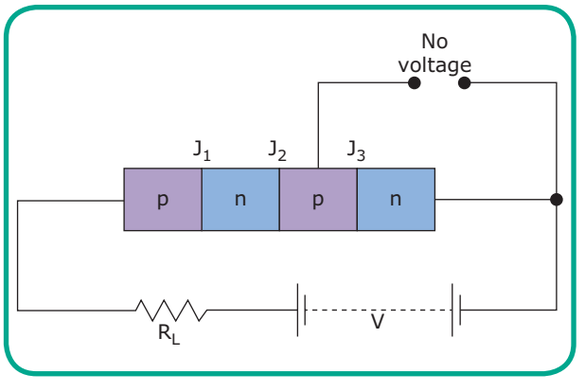


Figure 6.16 **Forward Biasing of SCR**

When supply voltage is gradually increased to a particular voltage called forward break over voltage, the reverse biased junction J_2 breaks down. The SCR now conducts heavily and is said to be in 'ON' state.

2. ***When Gate is applied positive voltage with respect to cathode***

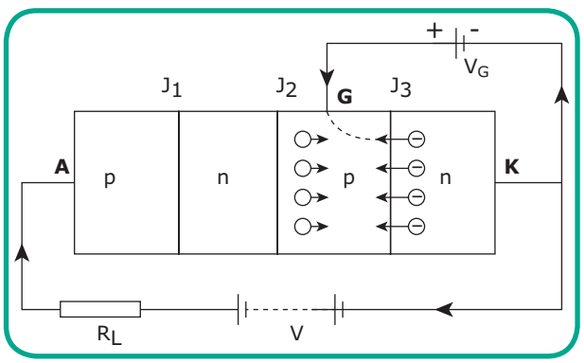


Figure 6.17 **SCR with Gate Voltage**

Figure 6.17 shows the small positive voltage applied to the gate. Now the junction J_3 is forward biased and junction J_2 is reverse biased. The electrons from N-layer (cathode) start moving across junction J_3 towards left, whereas holes from P-layer (Gate) move towards, right, consequently, the electrons from junction J_3 are attracted across junction J_2 and gate current starts flowing. This makes the anode current to increase. This increased anode current in turn makes more electrons available at junction J_2. This process continues and in short time the junction J_2 breaks down and SCR starts conducting heavily. Once the SCR starts conducting the gate loses its control. Even if gate voltage is removed, the anode current does not decrease at all. The only way to stop conducting is to reduce the supply voltage (V) to zero.

Application of SCR

1. The SCR is used in the circuit of AC voltage stabilizer.
2. It can be used as switch.
3. It is used in inverters.
4. It is used with AC power control with solid relay.
5. It is used to control motor speed.
6. It is used in light dimmer control circuits.

6.12. DIAC

The DIAC is a bidirectional semiconductor switching device. It can be switched 'ON' using both polarities. DIAC is a short version of DIODE Alternating Current. It is widely used as a triggering device of a Triac, especially, for AC switches, dimmer application and starter circuits in fluorescent lamps.

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